DATA DICTIONARY: Diabetes 130-US Hospitals (1999-2008)

Application of AI in Healthcare (MSAI-544) Residency - Group Project

DATASET INFORMATION

Source: UCI Machine Learning Repository (dataset 296), donated 2014 by the Center for Clinical and Translational Research, Virginia Commonwealth University

Original data: Cerner Health Facts, a national clinical data warehouse of de-identified electronic health records from participating US hospitals

Study: Strack et al. (2014), "Impact of HbA1c Measurement on Hospital Readmission Rates: Analysis of 70,000 Clinical Database Patient Records," BioMed Research International

Time period: 1999-2008 (ten years), 130 hospitals and integrated delivery networks

Size: 101,766 encounters, 71,518 unique patients, 50 columns

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Files:

- diabetic_data.csv (18.3 MB), one row per inpatient encounter
- ids_mapping.csv, descriptions for admission_type_id, discharge_disposition_id and admission_source_id (cleaned into one three-column table: id_type, id, description)

Inclusion Criteria (applied by the original authors)

1. Inpatient encounter (hospital admission)
2. Diabetes of any kind entered as a diagnosis
3. Length of stay 1 to 14 days
4. Laboratory tests performed during the encounter
5. Medications administered during the encounter

Loading

```
Python:  df = pd.read_csv("diabetic_data.csv", na_values=["?", "None"],
low_memory=False)
         ids = pd.read_csv("ids_mapping.csv", keep_default_na=False)
R:       df <- read_csv("diabetic_data.csv", na = c("?", "", "None"), guess_max
= 110000)
         ids <- read_csv("ids_mapping.csv", na = character())
```

"?" is the missing marker. "None" in A1Cresult and max_glu_serum means the test was not performed; treat it as missing but remember that "not tested" is itself informative. Two codes in ids_mapping have the literal description "NULL"; keep_default_na=False / na = character() stops the loaders from converting it to a missing value.

COLUMN DEFINITIONS

Identifiers

Column	Type	Description	Notes
encounter_id	Integer	Unique identifier of the hospital stay	Never use as a feature
patient_nbr	Integer	Unique identifier of the patient	71,518 patients; up to 40 encounters each. Rows of the same patient are not independent

Demographics

Column	Type	Description	Values / notes
race	Categorical	Race as recorded	Caucasian 75%, AfricanAmerican 19%, Hispanic 2%, Other 1.5%, Asian 0.6%, missing 2.2%
gender	Categorical	Gender as recorded	Female 54%, Male 46%, Unknown/Invalid (3 rows)
age	Categorical	Age band in 10-year intervals	[0-10) to [90-100); 70-80 is the largest band; convert to a midpoint for modeling
weight	Categorical	Weight band in pounds	97% missing; drop

Encounter Characteristics

Column	Type	Description	Values / notes
admission_type_id	Code (1-8)	How the patient was admitted	Emergency 53%, Elective 19%, Urgent 18%; codes 5, 6, 8 are unknown variants
discharge_disposition_id	Code (1-29)	Where the patient went at discharge	Home 59%, SNF 14%, home health 13%. Codes 11, 19, 20, 21 = Expired; 13, 14 = Hospice. Exclude these from a readmission cohort
admission_source_id	Code (1-25)	Where the patient came from	Emergency room 56%, Physician referral 29%, transfers
time_in_hospital	Integer	Length of stay in days	1 to 14, median 4
payer_code	Categorical	Primary payer	40% missing; MC (Medicare) 32%, HM, SP (self-pay), BC, MD (Medicaid), 13 others
medical_specialty	Categorical	Specialty of the admitting physician	49% missing; 72 values; InternalMedicine, Emergency/Trauma, Family/GeneralPractice, Cardiology most common.

Column	Type	Description	Values / notes
			Group into top-N + Other + Missing

Utilization (counts)

Column	Type	Description	Range (median)
num_lab_procedures	Integer	Lab tests performed during the encounter	1-132 (44)
num_procedures	Integer	Non-lab procedures during the encounter	0-6 (1)
num_medications	Integer	Distinct generic medications administered	1-81 (15)
number_outpatient	Integer	Outpatient visits in the year before the encounter	0-42 (0)
number_emergency	Integer	Emergency visits in the year before the encounter	0-76 (0)
number_inpatient	Integer	Inpatient admissions in the year before the encounter	0-21 (0). The strongest single predictor of readmission
number_diagnoses	Integer	Diagnoses entered for the encounter	1-16 (8)

Diagnoses

Column	Type	Description	Notes
diag_1	ICD-9 code	Primary diagnosis	716 distinct codes; 21 missing. Group into clinical categories (see below)
diag_2	ICD-9 code	Secondary diagnosis	748 distinct; 358 missing
diag_3	ICD-9 code	Additional secondary diagnosis	789 distinct; 1,423 missing

ICD-9 grouping used by Strack et al. (function provided in the starter code): Circulatory 390-459 and 785; Respiratory 460-519 and 786; Digestive 520-579 and 787; Diabetes 250.xx; Injury 800-999; Musculoskeletal 710-739; Genitourinary 580-629 and 788; Neoplasms 140-239; everything else (including V and E codes) Other.

Laboratory Results

Column	Type	Description	Values
max_glu_serum	Categorical	Highest glucose serum test result	>200, >300, Norm, or None (not tested; 95%)
A1Cresult	Categorical	HbA1c test result (average glucose over about 3 months)	>8 (poor control), >7, Norm, or None (not tested; 83%). The original paper

Column	Type	Description	Values
			studied whether ordering this test relates to readmission

Medications (23 columns)

For each generic diabetes drug, whether it was prescribed and whether the dose changed during the encounter. Values: **No** (not prescribed), **Steady** (prescribed, dose unchanged), **Up** (dose increased), **Down** (dose decreased).

Columns: metformin, repaglinide, nateglinide, chlorpropamide, glimepiride, acetohexamide, glipizide, glyburide, tolbutamide, pioglitazone, rosiglitazone, acarbose, miglitol, troglitazone, tolazamide, examide, citoglipton, insulin, glyburide-metformin, glipizide-metformin, glimepiride-pioglitazone, metformin-rosiglitazone, metformin-pioglitazone.

examide and citoglipton are "No" in every row; drop them. insulin is the most informative (No 47%, Steady 30%, Down 12%, Up 11%). Useful derived features: number of diabetes medications prescribed, number with a dose change.

Summary Flags and Target

Column	Type	Description	Values
change	Binary	Any change in diabetes medication (dose or generic name)	Ch 46%, No 54%
diabetesMed	Binary	Any diabetes medication prescribed	Yes 77%, No 23%
readmitted	Target	Days to inpatient readmission	<30 (11.2%), >30 (34.9%), NO (53.9%). Model <30 vs everything else

ID CODE REFERENCE (ids_mapping.csv)

id_type	Codes worth knowing
admission_type_id	1 Emergency, 2 Urgent, 3 Elective, 4 Newborn, 5 Not Available, 6 NULL, 7 Trauma Center, 8 Not Mapped
discharge_disposition_id	1 Discharged to home; 2 another short-term hospital; 3 SNF; 4 ICF; 5 another inpatient institution; 6 home with home health; 7 left AMA; 8 home under Home IV; 9 admitted as inpatient; 11 Expired; 13 Hospice/home; 14 Hospice/medical facility; 18 NULL; 19, 20, 21 Expired (variants); 22-30 transfers to rehab, long-term care, psychiatric, federal facilities, etc.
admission_source_id	1 Physician referral, 2 Clinic referral, 3 HMO referral, 4 Transfer from hospital, 5 Transfer from SNF, 6 Transfer from another facility, 7 Emergency room, 8 Court/law enforcement, 9 Not available, 17 NULL, 20 Not mapped, others are rare transfers and newborn sources

DERIVED FIELDS (TO CREATE)

Field	Formula	Purpose
readmit_30	readmitted == "<30"	Binary target
age_num	Lower bound of age band + 5	Numeric age (band midpoint)
diag_group	icd9_group(diag_1)	Clinical category of primary diagnosis
total_prior_visits	outpatient + emergency + inpatient	Overall prior utilization
n_diabetes_meds	Count of medication columns != "No"	Treatment intensity
n_meds_changed	Count of medication columns in {Up, Down}	Regimen instability
a1c_tested, a1c_high	A1Cresult not missing; A1Cresult in {>7, >8}	Testing behavior and glycemic control
discharge_group	Home / Home with services / Facility transfer / Other	Collapses 26 observed codes
source_group	Emergency room / Referral / Transfer / Other	Collapses 17 observed codes

DATA QUALITY ISSUES

1. Repeated patients

30% of encounters belong to patients who appear more than once. Encounters of the same patient share demographics, history and often outcomes. If they are split across training and test sets, the test estimate is optimistic. Either keep one encounter per patient (the original study kept the first) or split by patient.

2. Outcomes that cannot occur

About 2% of encounters ended in death or discharge to hospice. These patients cannot be readmitted; leaving them in teaches the model that "Expired" predicts "not readmitted," which is both useless and a form of leakage.

3. Informative missingness

Missing A1Cresult means the test was not ordered, which the original authors found was itself associated with readmission. Missing payer_code and medical_specialty are also patterns, not random gaps. Encode missing as a category rather than dropping rows.

4. High-cardinality codes

Over 700 distinct diagnosis codes, 72 specialties, 26 discharge codes. One-hot encoding them raw creates thousands of sparse columns and overfits. Group them.

5. Class imbalance

About 9% of the cleaned cohort is readmitted within 30 days. Accuracy is meaningless (predicting "no" for everyone scores 91%). Use ROC AUC, PR AUC, calibration and sensitivity at a stated operating point.

6. Age of the data

Collected 1999-2008, before the Affordable Care Act, the Hospital Readmissions Reduction Program (2012), ICD-10 and most modern diabetes drugs. Patterns are real but may not transfer to today's hospitals.

7. What is not in the data

No social determinants (housing, income, caregiver support), no medication adherence, no vital signs or lab values beyond two categorical results, no post-discharge follow-up information, no hospital identifier. These are among the strongest known drivers of readmission, which caps achievable model performance.

CLINICAL AND POLICY CONTEXT

Why 30-day readmission?

An unplanned return to hospital within 30 days is widely used as a quality measure because many such readmissions reflect problems with discharge planning, medication reconciliation or follow-up care rather than new disease. Since 2012 the US Centers for Medicare and Medicaid Services has penalized hospitals with excess readmissions for selected conditions under the Hospital Readmissions Reduction Program, with penalties of up to 3% of Medicare inpatient payments.

Diabetes and readmission

Patients with diabetes are readmitted at higher rates than patients without it (roughly 14-22% vs 9-13% at 30 days in published series). Contributing factors include multiple chronic conditions, complex medication regimens, hypoglycemia after discharge, and gaps in self-management support.

What a risk model is for

Transitional-care interventions (pharmacist medication review, nurse phone calls, early follow-up appointments, home visits) reduce readmissions but are labor-intensive. A risk model lets a hospital aim them at the patients most likely to benefit. The practical question is not "is the AUC high" but "if we can reach 15% of discharges, which 15% should it be, and how many readmissions does that prevent?"

Realistic performance

Published readmission models, including those on this dataset, typically reach ROC AUC of 0.60-0.70. The LACE and HOSPITAL scores used in practice are in the same range. A model near 0.65 that is well calibrated and fairly applied is operationally useful; a model near 0.90 on this data is almost certainly leaking.

Fairness considerations

Readmission models have been shown to under-flag some groups because the data encode unequal access to care (for example, fewer recorded prior visits for patients with less access). Differences in sensitivity or flag rate across race, sex and age at the operating point you would actually use are the relevant test, not overall AUC alone.

EVALUATION METRICS REFERENCE

Metric	Meaning	Why it matters here
ROC AUC	Probability a random readmitted patient scores higher than a random non-readmitted one	Threshold-free discrimination; the standard headline number
PR AUC (average precision)	Area under precision-recall; baseline equals the positive rate (about 0.09)	More informative than ROC AUC when positives are rare
Brier score	Mean squared error of predicted probabilities	Overall probability quality; lower is better
Calibration curve	Predicted probability vs observed rate by decile	A risk score clinicians can trust must mean what it says
Sensitivity at operating point	Share of readmissions flagged when the top X% are flagged	What the outreach team actually captures
PPV at operating point	Share of flagged patients who are readmitted	How many calls are "wasted"
Number needed to flag	Patients flagged per readmission captured	Workload per potentially prevented event
Cumulative gains	Share of readmissions captured vs share of patients flagged	Capacity planning curve

CITATIONS

Clore, J., Cios, K., DeShazo, J., & Strack, B. (2014). Diabetes 130-US Hospitals for Years 1999-2008 [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5230J

Strack, B., DeShazo, J. P., Gennings, C., Olmo, J. L., Ventura, S., Cios, K. J., & Clore, J. N. (2014). Impact of HbA1c measurement on hospital readmission rates: Analysis of 70,000 clinical database patient records. BioMed Research International, 2014, 781670. https://doi.org/10.1155/2014/781670

GLOSSARY

- **Encounter:** one hospital admission; the unit of analysis in the raw file
- **HbA1c:** glycated hemoglobin, a lab test reflecting average blood glucose over about three months
- **ICD-9:** the diagnosis coding system used in the US until 2015
- **SNF / ICF:** skilled nursing facility / intermediate care facility
- **AMA:** against medical advice
- **Transitional care:** services that bridge discharge and the next care setting
- **Operating point:** the probability threshold (or top-X% rule) at which a score triggers action
- **Leakage:** training on information that would not be available at prediction time or that encodes the outcome
- **Calibration:** agreement between predicted probabilities and observed rates
- **Model card:** a short standardized document describing a model's intended use, performance, limitations and governance

Questions about the data? Ask your instructor during residency work time.